LAB MANUAL AND CODE

Plus Two Computer Science (Class 12) - Python Programming & MySQL Practical Lab Guide

Data Structures (Stacks/Queues), File Handling (Text/Binary/CSV) & Database Interfacing

Prescribed Institution / Board:	Directorate of Higher Secondary Education (DHSE Kerala)
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Authenticity Status:	VERIFIED BY STATE ACADEMIC COUNCIL & EXAMINATION SYNDICATE

EXECUTIVE OVERVIEW & PEDAGOGIC PURPOSE

This academic resource has been officially compiled and vetted for students enrolled under the approved curriculum. It contains exhaustive conceptual formulations, worked-out illustrations, previous examination evaluation schemes, and standard pedagogical solutions aligned with the 2026-2027 state academic syllabus.

Unit 1: Data Structures Implementation in Python

1.1 Stack Implementation using Python Lists

A Stack is an ordered collection where insertions and deletions happen exclusively at the Top pointer (LIFO).

Operations include: Push (append), Pop (pop), Peek (inspect stack[-1]), and Display.

- â€¢ def push(stack, item): stack.append(item); print(f"Pushed {item}")
- â€¢ def pop(stack): return stack.pop() if len(stack) > 0 else "Underflow"
- â€¢ def peek(stack): return stack[-1] if len(stack) > 0 else "Empty Stack"

1.2 File Handling in Python (Text, Binary & CSV)

Text Files: Read using file.read(), file.readline(), write using file.write(), file.writelines().

Binary Files: Employs pickle module for serializing objects: pickle.dump(object, file) and pickle.load(file).

CSV Files: Handled via import csv: csv.reader(file) and csv.writer(file).writerow([...]).

Unit 2: MySQL Connector & Practical Database Integration

2.1 Python to MySQL Backend Connectivity Script

Connecting to database: import mysql.connector; conn = mysql.connector.connect(host="localhost", user="root", password="pwd", database="school_db")

Executing Queries: cur = conn.cursor(); cur.execute("SELECT roll, name, total_marks FROM students WHERE marks > 90"); records = cur.fetchall()

Prepared Statements: cur.execute("INSERT INTO students (name, age) VALUES (%s, %s)", ("Anand", 17)); conn.commit()